

Basketball

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Performance Demands of Basketball

In this section, we introduce the sport of basketball, and discuss the various performance factors and the relative contribution that physical performance makes toward achieving performance and winning.

Sport Description and Factors of Winning

Basketball is a stochastic, intermittent high-intensity sport characterized by periods of high aerobic oxidative and anaerobic glycolytic demands, continuous changes of direction (COD) that challenge the neuromuscular system, including accelerations and decelerations, jumps, sprints, physical contact, and sport-specific skills (7, 28). Furthermore, cognitive demands, such as perception, decision-making, and anticipatory processes are considered key winning factors in basketball (1, 7, 38, 48). The players are typically classified into five playing positions that include point guard (PG), shooting guard (SG), small forward (SF), power forward (PF), and center (C), although it is also common to pool players into just three categories of guards (G), wings (W), and bigs (B). Each of these playing positions requires specific skills, along with differing physical performance demands.

Depending on the body that regulates the competition (e.g., International Basketball Federation [FIBA], National Basketball Association [NBA], Asociación de Clubes de Baloncesto [ACB], National Collegiate Athletic Association [NCAA], etc.), there are small differences in terms of playing times and court dimensions. In the NBA, for instance, the game consists of four periods of 12 min, played on a 28.7 by 15.2 m (94 × 50 ft) indoor court. FIBA prescribes a game consisting of four periods of 10 min, with indoor court dimensions of 28 by 15 m (91.9 × 49.2 ft). The team that scores the most points by the end of regulation time is declared the winner.

Relative Contribution of Physical Performance

Basketball players require high levels of physical conditioning to elicit technical and tactical skills throughout a game (figure 23.1). Although all three aspects of speed, strength, and endurance are relevant to the player, the percentage contribution of importance for any given game will depend on the characteristics and style of the player (e.g., a tactical point guard, a three-point shooting specialist guard, a physical center) and the configuration of the roster, which may be more focused on emphasizing any three of these aspects. The desired performance-based characteristics of a basketball player are simply to:



Figure 23.1 The position of the basketball player on the three axes illustrates the relative importance of the three main physical capacities of importance for elite participation in basketball, acknowledging this varies slightly between positions. Although all three aspects are relevant in the game, the percentage of participation of these aspects varies depending on the characteristics of the player and the team. The pie chart shows the balanced relative importance of skills (33%), tactical awareness (33%), and physical capacities (33%) for team success, again acknowledging the variance across positions.

Adapted from G.A. Nader, "Concurrent Strength and Endurance Training: From Molecules to Man," *Medicine & Science in Sports & Exercise* 38, no. 11 (2006): 1965-1970.

1. run faster and jump higher than the opponent,
2. have strength and balance to endure physical contact and collisions involved in the game, and
3. perform these demands with less fatigue than the opponent.

Furthermore, these tasks must be carried out optimally in relation to a specific context, that is, in coordination with teammates, against opponents, and according to ball location on the court.

Decision-making processes and expertise are additionally critical to maximize the player's overall athletic performance. A recent study (40) on game performance indicators comparing "all-star" versus "non-all-star" players in the NBA (1,230 games analyzed during the 2013-14 regular season) showed a tendency for all-star players to cover slightly less distance per game and at lower average velocities. This highlights the higher efficiency of these players versus non-all-stars, a finding that is consistent with others (55). Thus, all-star players appear to make fewer mistakes when deciding when and where to run in both offensive and defensive situations, possibly taking shorter paths to reach destinations. These fewer mistakes made could lead to lower total dis-

tances covered by these players, leading to lower energy demands during a game. In fact, research suggests that motor efficiency achieved through intensive training leads to improved perception, focus, anticipation, planning, and fast responses (57). In such specific environments, optimal actions do not necessarily require the peak potential of the player. Nevertheless, the better the potential of the individual, the greater the availability of resources, or lower relative cost of actions (44). All this together emphasizes the importance of proper exercise design, including how HIIT fits into the program.

Targets of Physical Performance in Basketball

Previous analysis of the physiological determinants of success in basketball has shown the importance of both aerobic oxidative and anaerobic glycolytic capacities (15). Due to the large number of short, high-intensity actions and basketball-specific movements, neuromuscular power capabilities, including accelerations and COD, screening, and

blocking or positioning for rebounds are equally important. A proper conditioning program must allow players to obtain, maintain, or enhance these physical capabilities, which ultimately may optimize sport performance while lowering the risk of injury.

Players

In a sport where the scoring goal is at 3.05 m (10 ft), a player's height is obviously a key factor for success. Thus, in basketball, anthropometry is of great importance. The adage “one size does not fit all,” despite being true across sport, is especially relevant in basketball, which has a large variance of size within any squad (36). In the NBA, the average height for PG is 1.86 m (6.1 ft), for W (SG and SF) 1.97 m (6.5 ft), and for B (PF and C) 2.06 m (6.7 ft). At the extremes, there are players as tall as 2.30 m (7.5 ft) (i.e., Manute Bol) or as short as 1.57 m (5.15 ft) (i.e., Muggsy Bogues). For most team sports, physical and physiological demands are position dependent in terms of movement patterns, distance covered, high-intensity actions, and work:rest ratios (7, 8, 45, 53). One of our group's last studies with Spanish professional basketball players (ACB league) (45) showed that guards reached the highest acceleration loads, irrespective of the drill performed. These results support two logical principles:

1. The smaller the player, the lower the body mass, with resulting easier accelerations due to less applied force (force = mass $\times$ acceleration; acceleration = force $\times$ mass⁻¹).
2. The tactical principles of basketball usually imply that the playing zones for bigger players are smaller in area than those for small players, meaning that small players usually have to cover more distance per play or possession for tactical reasons (45).

With regard to principle 1 (player size), it seems reasonable to use a scaling factor such as body mass or body mass index to minimize such inter-player differences and to obtain an individualized external load. As previously reported in other team sports, such as Australian football, the identification of position-specific acceleration profiles would assist coaches and staff members, as well as sports scientists, to develop position-specific dependent drills aimed at improving a player's conditioning (54).

Key Weapons, Manipulations, and Surveillance Tools

Our previously proposed conditioning training methodology (43, 44) is based on:

- exercise specificity progression according to the task orientation (how similar the exercise is in relation to an actual basketball game: general, directed, special, and competitive) (47),
- the approach level (0⁻, 0⁺, I, II, III, IV, and V; described further below) related to the orientation (31), and
- player needs.

Using this as a framework, the following high-intensity interval training (HIIT) weapons form part of the training progression (from general to specific). Based on the physical, physiological, and cognitive targets, we manipulate each of the variables defining our conditioning sessions (i.e., chapter 4), including work and rest modality, intensity, duration, number of sets and repetitions, as well as decision-making complexity, and others (10, 28, 44).

HIIT Weapons

In our opinion, well-designed general aerobic power development using long-interval HIIT (level I; types 2 and 3) should be carried out in the off-season, early in the preseason, or to address specific player needs. During the preseason, we recommend focusing on game-based HIIT (GBHIIT) or basketball-based repeated high-intensity effort ability using small-sided games (SSGs) and actual basketball games mainly (levels IV and V, respectively). Eventually, we may also want to include less cognitively demanding but neuromuscularly challenging methods such as run-based short HIIT, repeated-sprint training (RST), or sprint interval training (SIT) with COD (types 3, 4, and 5), but we would rather integrate them as part of basketball-based drills manipulating their constraints (levels II and III). The progression of volume and intensity depend on the players' characteristics and fitness levels, but ideally, we will build up the players' capability to cope with high-intensity loads throughout the preseason to prepare them for the high demands of competition. During the season, skill-based conditioning and SSGs tend to predominate (levels IV and V, types 3 and 4), but we should not lose sight of the risk involved in losing control of level V work. Thus, the periodic

Table 23.1 Conditioning Specific Training: Orientation and Approach Level Characteristics

Orientation	Approach level	Similarity	Training method	Intensity	Confrontation format	Place	Ball
Competitive	V	Basketball	Actual game	Optimal	5 v 5	On court	With ball
			Simulated game	Supra-, equal-, or infra-competitive	5 v X, 4 v 4		
Special	IV	Basketball	Small-sided games	Supra-, equal-, or infra-competitive	4 v X	On court	With ball
					3 v 3, 3 v X		
					2 v 2, 2 v X		
					(1 v 1)		
Directed	III	Basketball-based	RST (COD) (<7 s)	All-out or supra-competitive	1 v 0 (X v 0)	On court	With/without ball
			SIT (COD) (7-15 s)				
	II		Short HIIT (COD) (15-60 s)	Anaerobic speed reserve (ASR)			
General	I	Run-based/ basketball-based	Short HIIT (15-60 s)	Anaerobic speed reserve (ASR)	None/X v 0	Off/ on court	With/without ball
			Long HIIT (> 60 s)	>90% $\dot{V}O_2$ max			
	0 ⁺	Nonspecific (run-based)	Continuous or interval training	<85% $\dot{V}O_2$ max	None	Off court	Without ball
	0 ⁻	Nonspecific					

/: optional; (): optional but normally unused; X: a number smaller than the first number (e.g., 3 v X=3 v 1 and 3 v 2, but not 3 v 3 or 3 v 4); ASR: anaerobic speed reserve, faster than $\dot{V}O_2$ max speed and slower than maximum sprint speed; $\dot{V}O_2$ max: maximal oxygen uptake; min: minutes. Here are shown just a few examples; there are myriad options.

* As defined by Buchheit and Laursen, 2013.

Adapted by permission from X. Schelling and L. Torres-Ronda, "Conditioning for Basketball: Quality and Quantity of Training," *Strength and Conditioning Journal* 35 no. 6 (2013): 89-94.

Orientation	Type 1 HIIT* Metabolic (O ₂)	Type 2 HIIT* Metabolic (O ₂) + neuromuscular	Type 3 HIIT* Metabolic (O ₂) + La ⁺	Type 4 HIIT* Metabolic (O ₂) + La ⁺ + neuromuscular	Type 5 HIIT* Metabolic (peripheral O ₂) + La ⁺ + neuromuscular
Competitive	The characteristics of an actual game imply all five types of physiological demands depending on the player's playing position and pacing strategy and the context of the game (schedule, score, remaining time, opponent characteristics, etc.)				
		4v4-4v4-4v4 in <15 s, (3 courts, ~40+40 s), 3 groups, 2:1	4v4-4v4-4v4 in < 15 s, (3 courts, ~40 s), 3 groups + speech, 1:1	5 v 4(+1) in < 10 s, (3 courts, ~30 s), 2 groups + speech, 1:2	
Special		4v3-3v2-2v1 (3 courts, ~30+30 s), 3 groups, 2:1	4v3-3v2-2v1 (3 courts, ~30 s), 3 groups + speech, 1:1	4v3-3v2-2v1 (3 courts, ~30 s), 3 groups + speech, 1:2	
		3 v 3 full court (2 courts, ~20+20 s), 4 groups, 2:1	3 v 3 (6 v 6) (2 courts, ~20 s), 2 groups, 1:1	Fast break 3v2-2v1 (2 courts, ~15 s), 3 groups + speech, 1:3	
		2 v 2(+2p) (2 courts, ~15+15 s), 3 groups, 2:1	2 v 2(+2p) (2 courts, ~15 s), 3 groups + speech, 1:1	Fast break 2v1-3v0 (2 courts, ~20 s), 4 groups of 3, 1:3	
		2-way 1 v 1 defense, (2 courts, ~8+8 s), 6 groups, 2:1	1 v 1 defense, (1 court, ~8 s), 6 groups + speech, 1:1	1 v 1 full court defense, (1/2 court, ~5 s), 6 groups of 2, 1:4	
Directed				All-out or supra-competitive skill-based session (1 v 0), 2-5 s with incomplete recovery	All-out or supra-competitive skill-based session (1 v 0), 2-5 s with complete recovery
					All-out or supra-competitive skill-based session (1 v 0), 10-15 s with complete recovery
		Fast break 3 v 0 (3 courts, ~20 s), 2 groups of 6, 2:1	Fast break 4 v 0 (3 courts, ~20 s), 2 groups of 4, 1:1	Fast break 3 v 0 (3 courts, ~20 s), 4 groups of 3, 1:3	
General		2 × [8 × (15 s on/15 s off)] at ASR+ or 2 × [8 × (30 s on/30 s off)] at ASR+	2 × [8 × (15 s on/30 s off)] at ASR++ or 2 × [8 × (30 s on/60 s off)] at ASR++	2 × [8 × (15 s on/45 s off)] at ASR+++ or 2 × [8 × (30 s on/90 s off)] at ASR+++	
			6 × 2 min at 90%-95% $\dot{V}O_2$ max, 1 min rest	6 × 2 min at 95%-100% $\dot{V}O_2$ max, 2 min rest	
	Not necessarily involving HIIT, but it has to be run-based (e.g., 3-4 × 8 min at 75%-80% $\dot{V}O_2$ max, 2 min rest)				
	Not necessarily involving HIIT. This level is meant for non-sport-specific, low-intensity, longer sessions such as long riding, rowing, or swimming at low-moderate intensity (e.g., 30 min at 65%-70% $\dot{V}O_2$ max).				

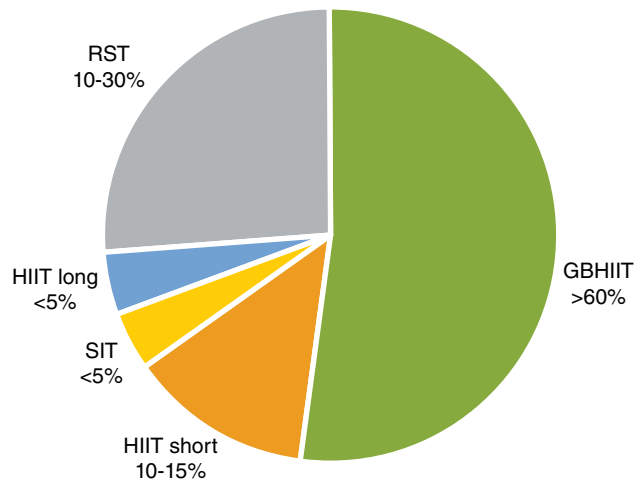


Figure 23.2 Percentage of the different HIIT weapons (formats) used throughout the annual season in elite basketball. HIIT: high-intensity interval training; RST: repeated-sprint training; SIT: sprint interval training; GBHIIT: game-based HIIT. If we consider all individual development sessions skill focused, as SIT and RST with change of direction, the percentage of this HIIT weapon would be on the top range (30%), since the players tend to do these types of sessions almost every day (please see table 23.1). GBHIIT in the form of small-sided and official games are the main technical, tactical, and physical resources.

implementation of level II and/or level III work is important (e.g., once a wk or every 2 to 3 wk), especially with the players not playing minutes (table 23.1).

Training Manipulation

As discussed above, we approach the conditioning training (obtain, maintain, or enhance) based on the principles of sport specificity, task orientation, and player needs. Once we know the needs of the individual player and the main goals of the team, the main aim of the training session (metabolic, neuromuscular, cognitive) becomes established, and the principle of specificity guides us in choosing the best weapon to implement (i.e., SIT, RST, short HIIT, long HIIT, SSG, or scrimmage; figure 23.2). The periodization, either at a weekly or monthly level, of each of these weapons is not discriminant nor conducted in isolation, but is more an overlap of the different training levels desired, while respecting the much needed recovery processes, high-intensity tolerance, and workload variety. Hereafter we describe our training orientation model, including its general, directed, special, and competitive appli-

cations, alongside the HIIT weapons and target types referred to throughout the book.

General Orientation (Levels 0⁻, 0⁺, and I)

The general orientation levels are associated with aerobic and anaerobic responses. Unless a player has special needs (e.g., limiting injuries, joint or tendon pain, etc.), continuous moderate intensity run, bike, and swim training is not a priority conditioning task for basketball players. In fact, we prefer using HIIT as soon as possible. The levels 0⁻ and 0⁺ can be performed across any exercise mode (running, cycling, swimming, rowing, etc.) and the maximum training volume is determined either by a player's distance covered in a game (<2000-5500 m (<1.24-3.42 mi (8, 40))) or total game duration (<42 min of game time or <2 h of real time). The intensity varies depending on physiological targets (aerobic efficiency, aerobic capacity, or aerobic power). In our opinion, the most important goal is to improve the maximal oxygen uptake ($\dot{V}O_{2max}$) characterized by lower volume and higher training intensities (see chapter 3). It is important to consider that team sport athletes require a fairly high level of aerobic fitness in order to generate and maintain power output during repeated high-intensity efforts and to optimize their recovery (49). At this level, physical activity can be either sport specific or not, including varying surface types such as grass, the basketball court, with or without the ball, with or without specific movement patterns or specific skills, etc. The only critical aspect that should be respected is that the proposed exercises (movement patterns, technical skills, etc.) should not limit the required metabolic responses needed to target the training goal.

- **Level 0⁻:** Non-sport-specific physical activity. Decision-making does not appear at this level. Here, we suggest using different activities that do not involve specific movement patterns, such as cycling, swimming, rowing, etc., enabling low joint impact, and not necessarily involving HIIT (e.g., as a recovery workout (50) or conditioning injured players (11)).
- **Level 0⁺:** Non-sport-specific physical activity but using similar locomotor responses as basketball. While movement patterns here are not sport specific, activities are primarily run based. Decision-making does not come into play, but intensity is higher than level 0⁻ and HIIT with long intervals is recommended (type 2),

although continuous methods can also be used at this level.

- **Level I:** Actions and movement patterns replicate those seen in basketball. Decision-making does not come into play or is very simple and nonspecific and not a limiting factor to attaining the metabolic and neuromuscular targets. Here, we can work with basketball skill-based drills, using already consolidated skills (e.g., shooting drills or X v 0 half or full court; note: X implies any number of players, 0 implies no opposition, v implies versus). If the player's basketball skills are not yet well developed (e.g., junior players), it may be better to use generic drills (running based with basic basketball-movement patterns), either on or off the court. Intensity should range from 90% to 100% of $\dot{V}O_{2\max}$ (29), generally using long-interval HIIT, >60 s (type 3) (4, 10, 16, 41).

Directed Orientation (Levels II and III)

These orientation levels refer to all-out efforts, including RST and SIT, as well as very-high but not all-out short efforts (short HIIT), aiming for type 4 and 5 targets (10). These drills are performed on court and include COD and basketball-specific formats, taking into account playing-position characteristics. For us, useful drills include tactical situations with specific pathways, involving offensive or defensive plays, etc. We also recommend performing these exercises without opposition (1 v 0, 2 v 0, 3 v 0), to allow maximal intensity. The two directed orientation levels are described:

- **Level II:** Drills simulate basketball movement patterns and context. Decision-making is simple and basketball based or it is absent (e.g., X v 0). The intensity of the effort is very high but not maximal (within anaerobic speed reserve (ASR) (10)). This level is physiologically associated with high anaerobic metabolic and neuromuscular contributions. Drills are performed on court, using HIIT with short intervals (e.g., 15 s on/40 s off, ASR efforts interspersed with 2 to 4 min passive recovery periods; work:rest ratio of 1:3).
- **Level III:** Drills simulate basketball movement patterns and context. Decision-making is simple and basketball based, although it may be absent (e.g., X v 0). Drills must allow maximal effort. This level targets acceleration and deceleration abilities using RST or SIT methods, defined by

the recovery type (complete or incomplete) and the duration of the effort (<7 s or 7-15 s) (see table 23.1 for examples). Due to the court dimensions, linear maximal efforts longer than 5 or 6 s are rare in basketball, hence including specific COD and jumps helps lengthen efforts more appropriately.

Special Orientation (Level IV)

We consider the special orientation level essential for skill-based conditioning, in the form of SSGs (2 v 2, 2 v X, 3 v 3, 3 v X, and 4 v X) (types 2, 3, or 4; see table 23.1 for examples). Decision-making is complex and basketball specific. Several authors have argued that SSGs are as efficient as other forms of HIIT to develop specific aerobic and anaerobic characteristics for team sport players (2, 23, 25, 49), with benefits that include:

1. better transfer of physiological adaptations when the exercise simulates sport-specific movement patterns,
2. athletes simultaneously develop technical and tactical skills under high physical loads, and
3. motivation is higher for athletes performing sport-specific rather than traditional conditioning (2, 19, 27).

As described in chapters 2 and 5, however, controlling the overall load cannot be precisely standardized or, in our opinion, can be more complex. However, new technologies and tracking systems are allowing us to have a much better understating and control of these drills (45). Careful consideration of a player's skill levels, current fitness, number of players involved in the task, court dimensions, game rules, work:rest ratios, and availability of player encouragement is required (27, 35, 49). By manipulating these constraints, we can influence the overall physiological and cognitive workload (2). While somewhat described already in chapters 2 and 5, we would like to highlight the following training variables that are important for basketball SSGs:

1. **Number of players:** Reducing the number of players over the same court size results in increases in physiological demands (14). The 3 v 3 and 2 v 2 full-court (5) drills represent great formats for aerobic and anaerobic training stresses, when higher than (supra-) actual game conditions are desired (see table 23.1 for examples).

2. Court dimensions: Exercises can be classified by the number of courts required: half court (1/2; the drill is carried out only in half court), half court plus one court (1/2 + 1; the drill is carried out in half court, but includes a fast break or transition to the opposite basket), half court plus two courts (1/2 + 2; the drill is carried out in half court, but includes a fast break to the opposite basket, followed again by a transition finishing back to the same basket where the drill started), and open court (X courts; drills with more than two transitions or fast breaks such as 3, 4, 5, etc.). The topic of court dimensions is expanded on below.

3. Work:rest ratios: When we are designing any training exercise, it is important to know the characteristics of our sport and, accordingly, to propose training that is higher than (supra-), equal to (equal-) or less than (infra-), the actual game conditions. If the mean work:rest ratio for an actual game is about 1:4 (actually 1:3.6 (8)), we can manipulate this in training using player rotations (i.e., number of players playing on court and number of players resting off court) and modifying game rules (reducing or increasing the stop time by using fouls, out of bounds, free throws, etc.) (2, 20, 35). Within any drill design, we can use two different resting periods, termed intra-set and inter-set pauses. Intra-set pauses refer to the time that the drill design allows the player to rest within a set. Inter-set pauses refer to the time that the coach gives to the player to rest between different sets of the same exercise. These pauses, which determine the work:rest ratio, affect both the total and the relative load or intensity. For instance, with longer rests we can make the drill less lactic (change from HIIT type 4 to 2). Thus, when considering SSG drills, the workload, intensity, and its metabolic and neuromuscular responses must be considered, not only with the game format (1 v 1, 2 v 2, 3 v 3, 4 v 4, or 5 v 5) and playing space (half court, full court, open court), but also with rotations and associated intra- and inter-set pauses (45).

4. Coach encouragement: The external motivation provided by coach supervision has been shown to achieve greater gains and training adherence for SSGs and should be used as required (2, 35).

In light of these highlighted points, a good high-intensity exercise level in basketball tends to be achieved with 2 v X or 3 v X full-court drills, with coach encouragement throughout the drills. Limiting dribbling or possession time is also a useful means of increasing intensity.

Competitive Orientation (Level V)

The competitive orientation level is the most specific conditioning format, involving the most realistic physical, physiological, and cognitive requirements. The decision-making is complex and basketball specific. Exercises involve 4 v 4, 5 v X, and 5 v 5. The value of involving a larger number of players in these SSGs lies in enhancing team-specific decision-making skills: more teammates and adversaries are involved in the decision-making processes (17). We believe that in team sports, conditioning training is a way to improve player capabilities (fitness, cognition, technique, tactic, teamwork, etc.) but is never an exclusive goal. Players must be better at level V (playing actual basketball), and not, for example, at level 0+ or level III orientations. Nonetheless, training exclusively at levels IV and V is risky, because tasks are open, and some players might not receive enough of a needed training response/stimuli (due to insufficient effort or requirement in the level IV/V task), and as a result, lose fitness. In this sense, the topic “play as you train and train as you play” is crucial, meaning if your goal is that your team runs every fast break as fast as they can, fights for every ball, or collects every rebound, you must demand the same in every action of every drill to instill the desired attitude in your players. If not, their fitness level will be the least of your concerns. The design of drills at this level should follow the considerations discussed in the special orientation, which are common to SSGs. In this level it is usual to use game incentives (e.g., points) or modify other rules.

Metabolic Responses to Changes in Court Dimensions Several studies have shown that with the same number of players, increasing the court dimensions results in incremental physiological demands (i.e., full-court games produce significantly higher physiological responses than half-court games (5, 30, 45, 53), so that type 2 exercises become type 4. However, a smaller playing space entails significantly higher frequency of technical actions, and, consequently, more COD, accelerations and decelerations, etc. (12, 13, 38, 41).

In one of our recent studies with professional basketball players from the Spanish 1st division (ACB league) (53), the manipulation of court dimensions (half court, full court, and more than one court with different drills across X v 0, 1 v 1, 2 v X, 2 v 2, 3 v X, 3 v 3, 4 v X, 4 v 4, 5 v X, 5 v 5) showed that larger sizes led to a greater increase in HR, with the more-than-one court drills showing the higher HR values,

ranging from moderate to large effects. Previous studies have also shown that the fewer the players involved in the exercise within the same court size, the greater the HR_{avg} (14, 18, 39). Nevertheless, in order to properly assess the metabolic and neuromuscular demands of SSGs, research using a portable gas analyzer and EMG is needed due to the limitations that the HR response alone presents across variable and short high-intensity actions (9, 42).

Locomotor Responses to Changes in Court Dimensions

In a recent time-motion study on our group of professional basketball players, we found that even though there was a tendency for more frequency of movement per minute when the court was smaller (full-court versus half-court drills) these results were not constant across positions (G, W, and B), with each position showing different trends (53).

In another of our studies with professional basketball players (45), where we analyzed workloads with accelerometers, results showed that full-court 3 v 3 and 5 v 5 had the highest acceleration load per minute compared to other traditional balanced

basketball drills (2 v 2 and 4 v 4). Consistent with previous studies, 3 v 3 full-court drills seem to be a good drill choice for developing a higher game pace (supra-competitive drill), due to the smaller player number (13, 52). Interestingly, half-court 5 v 5 also showed high acceleration loads per minute, which may be due to its ecological validity, since the cognitive and physical requirements are closer to an actual basketball game, and the smaller size of the court may imply more interventions per player (37).

The following SSG examples (figures 23.3 to 23.8) run across different court sizes and confrontation formats for basketball. The drills include work:rest ratios and associated HIIT type. For more examples, and their acceleration loads, see Schelling and Torres-Ronda (45).

Load Surveillance and Monitoring Tools

It is important to take into account that, as of September 2018, in basketball, the use of wearables is not permitted during official games. Thus, as described

2 v 2 Fast break (6 v 6)

Number of players: 2 v 2

Space: full court (2 courts)

Bout duration (approximately): 15 to 20 s

Work:rest ratio (approximately): 1:2

HIIT type: type 4

Description

Player A attacks to player B and defends player C. Player B gets the rebound and passes to player C, who runs a fast break against player A. Rotation: A to B to C to A.

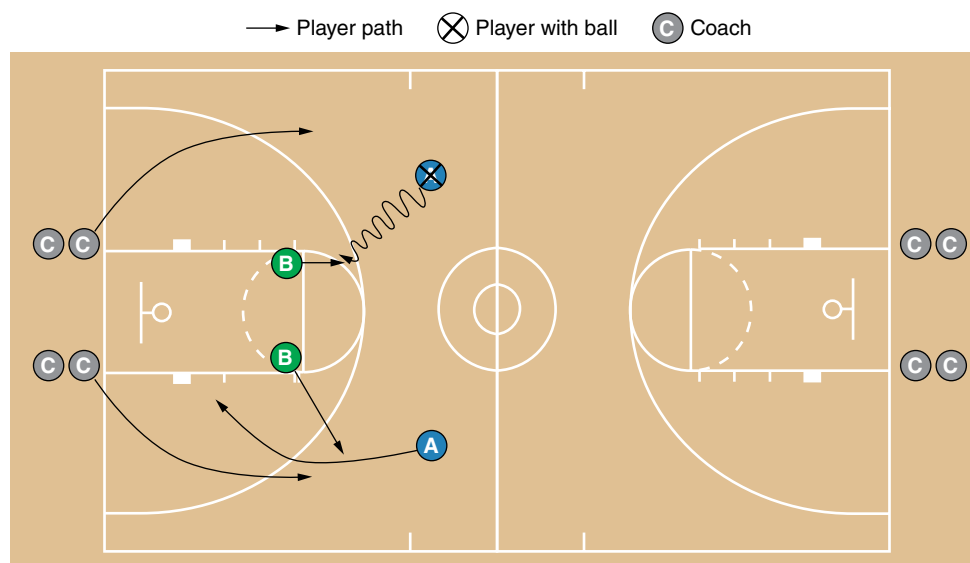


Figure 23.3 2 v 2 fast break (6 v 6).

3 v 3 Fast break (6 v 6)

Number of players: 3 v 3

Space: full court (2 courts)

Bout duration (approximately): 20 to 30 s

Work:rest ratio (approximately): 1:1

HIIT type: type 3

Description

Player B attacks to player A and defends player D. Player A gets the rebound and passes to player D who runs a fast break against player B.

Rotation: offense to defense to rest.

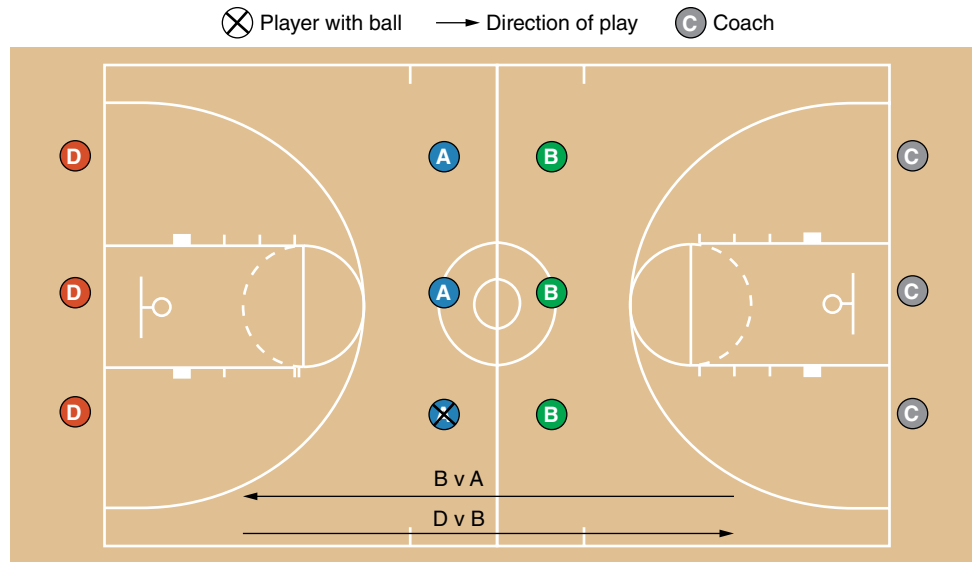


Figure 23.4 3 v 3 fast break (6 v 6).

in chapter 9, a disconnect is inevitable when assessing training and competition external and internal loads with the various monitoring tools at our disposal.

Nowadays, there is a large array of tools we can use to assess both the internal response and the external loads, both through subjective and objective measurements (chapters 8 and 9). Undoubtedly, the use of technology presents a great opportunity to obtain data in real time using devices that are less invasive, lighter, more precise, smaller, safer, and cheaper. The challenge for technology companies is to provide multiple, integrative, and synchronized sources of information that describe both the external and internal load of the athlete using the least invasive manner possible (51).

One of the most recent advances in assessing in-game basketball performance is through the use of optical player-tracking technology. This technology uses computer vision systems designed with algorithms capable of measuring the positions of players

with a sampling rate of around 25 frames per second (34). Of course, kinematic variables such as distance, velocity, or acceleration may be derived from these data, and sampling frequencies might improve in the future. Another approach that has gained acceptance within the area of indoor player tracking are the devices that incorporate microelectromechanical systems, gyroscopes, magnetometers, and accelerometers into a single player-worn unit (12). The devices utilize triaxial accelerometers that are not positional based, but movement based (anterior-posterior, medial-lateral, and vertical) (3). In recent times, there has been an increase in radio frequency identification-based technology for assessing indoor positional data. With these systems (only allowed in practice, not in official games) we can obtain descriptors of sport-specific activities, such as the type, frequency, and intensity of player movements (33, 45, 56). These performance variables represent duality of the performer and the environment in

4 v 4/4 v 4/4 v 4

Number of players: 4 v 4

Space: full court (2.5 courts)

Bout duration (approximately): 30 to 40 s

Work:rest ratio (approximately): 2:1

HIIT type: type 2

Description

Player A attacks to player B on half court and defends player B on transition defense. After

player B's fast break, player A counterattacks player B once more and player B performs transition defense. When player A versus player B is finished, player A stays in as a defender and player C comes in as the attacker.

Rotation: starting in offense to starting in defense to rest. Rotation work:rest ratio is 2:1.

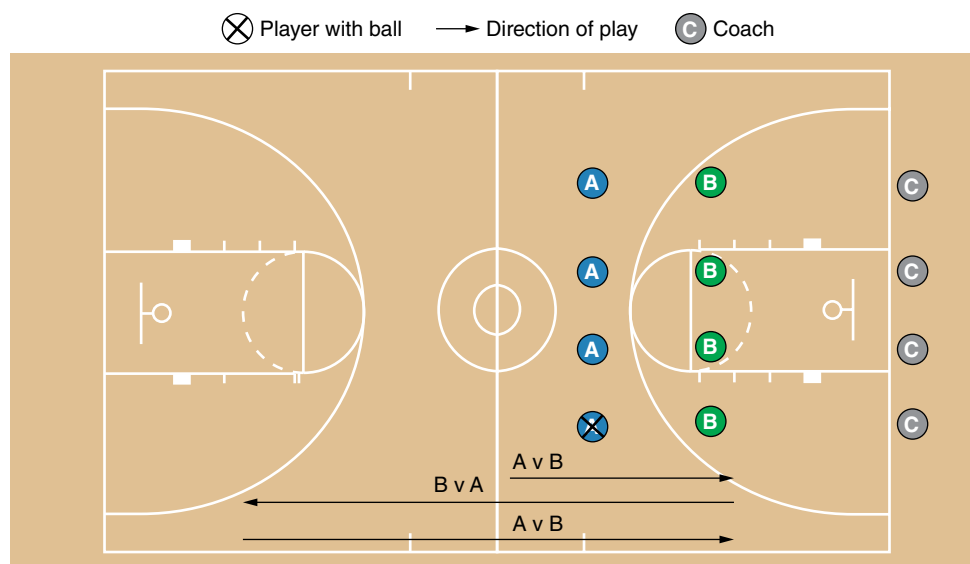


Figure 23.5 4 v 4/4 v 4/4 v 4.

order to understand how players engage with others alongside the resulting performance (21). In this sense, game- or practice-related statistics can provide insight into the function of the players as a successful unit. These measures of coadaptation describe the way that players function as part of a larger system (the team), coadapting to small but important changes in each other's structure and function (26).

For internal load, monitoring the HR response during training is the most common practice, but the use of small patches or wearables capable of measuring sweat composition, hydration level, glucose or lactate levels, among others, is also becoming more popular. As for subjective measurements such as rating of perceived exertion, they are affordable

and practical tools, but we also must keep in mind their limitations and validity (6, 32).

Training Status Surveillance and Monitoring Tools

Assuming that the total workload is the result of the combined psychological and biological demands (actual load) produced by training and/or competition (prescribed load) (22), which is additionally a function of the individual player characteristics (24), we must do our best to control the load appropriately to the actual state of our players. To do so, we recommend incorporating a variety of assessments, e.g., perceived effort or fatigue, specific biomarker levels, neuromuscular and power profiles, and HR response

5 v 5 (Half Court)

Number of players: 5 v 5

Space: half court

Bout duration (approximately): 10 to 24 s

Work:rest ratio (approximately): depends on coach's explanation

HIIT type: type 2 to 4; at higher pace with reduced rest, higher LA^+ ; at lower pace with reduced rest, higher O_2

Description

Player A attacks to player B (figure 23.6).

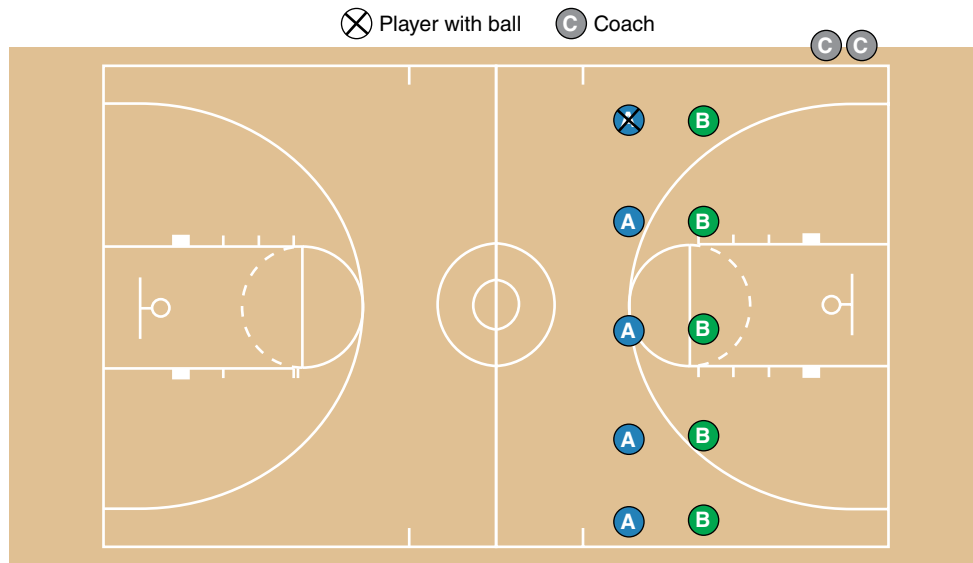


Figure 23.6 5 v 5 (half court).

to a standardized load, to track the acute and long-term adaptations to training and competition loads. Our personal experience with professional basketball teams has led us to perform daily, weekly, and monthly assessments that include self-reported wellness; maximal countermovement jump; specific biomarkers in blood, sweat, and saliva; heart rate response and recovery index in a submaximal test; and body composition. On top of this, additional assessments might be performed based on individual needs (acute or previous injuries, body composition, or basketball performance).

Strategies for Structuring the Training Program Using HIIT

In the following section we will outline the strategies we adhere to for structuring the training programming using HIIT, including the controllable and uncontrollable factors, the program structure and progression, alongside example programs.

Controllable and Uncontrollable Factors

As with most team sports, multiple games occur every week (1-5 games per wk). This schedule dictates the training and recovery prescription. This factor is uncontrollable, since it depends on the official body that organizes the competition (locally and internationally), and the only option is to adjust the training and recovery sessions accordingly.

A more controllable factor that can be managed is the minutes played per game (MPG). There are teams where the athletic performance department, with its sports scientists, doctors, physiotherapists, strength-and-conditioning coaches, etc., has some influence over in-competition player participation based on the individual accumulation and variance of external and internal loads. This scenario allows for a better periodization of training and recovery individually. On the contrary, some teams prescribe the in-game minutes only or base their decisions on

5 v 5/5 v 5/5 v 5

Number of players: 5 v 5

Space: full court (2.5 courts)

Bout duration (approximately): 30 to 40 s

Work:rest ratio (approximately): depends on coach's explanation

HIIT type: type 2 to 4; at higher pace with reduced rest, higher LA⁺; at lower pace with reduced rest, higher O₂

Description

Player A attacks to player B on half court and defends player B on transition defense (figure 23.7). After player B's fast break, player A counterattacks player B once more and player B goes into transition defense. When player A versus player B is finished, the resting time between sets depend on the coach's post-play feedback.

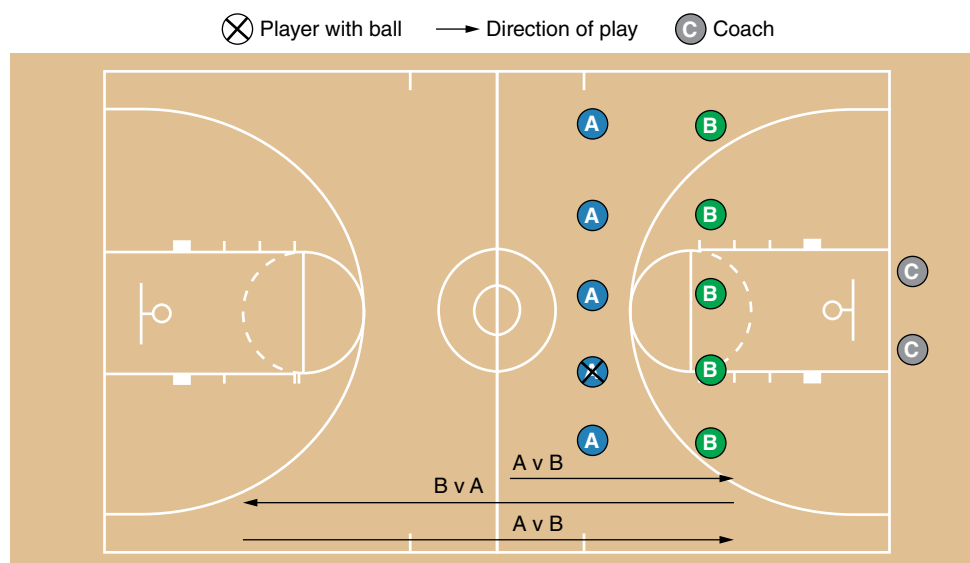


Figure 23.7 5 v 5/5 v 5/5 v 5.

a player's acute performance. The latter scenario requires one to compensate for the competition demands of each player. For example, some players play 35 min per game, three times a week, while other players play less than 10 min per game, once a week. Both groups of players require different strategies; one group needs adjustment for the lack of competitive stimuli, while the other group needs recovery for being exposed to very high workloads. This differentiated internal response, depending on the MPG throughout the season, was shown in a four-year study with professional basketball players that our group published in 2015 (46). This particular study clustered the players into three groups—less than 12 MPG, 13 to 25 MPG, and more than 26 MPG—with those players with less than 12 MPG and those with more than 26 MPG requiring special interventions; players with 13 to 25 MPG demon-

strated a pro-anabolic hormonal profile throughout the season.

Program Structure and Progression

From a periodization point of view, we divide the season into three main phases: off-season, preseason (open gym and training camp), and in-season. Off-season can range from several weeks to several months depending on the team. Thus, depending on individual player needs, this is the ideal, if not the only time, to apply training to maintain or enhance aerobic and/or anaerobic needs using more traditional training. During the preseason, we focus on improving the ability to repeat high-intensity efforts, interspersed with SSGs, actual basketball (levels IV and V), and official friendly games. During the in-season, the competition (in the NBA and for

5 v 5 Scrimmage

Number of players: 5 v 5

Space: full court (open court)

Bout duration (approximately): usually based on time (e.g., 3, 5, 6, 10 min)

Work:rest ratio (approximately): depends on coach's explanation

HIIT type: type 2 to 4; at higher pace with reduced rest, higher LA^+ ; at lower pace with reduced rest, higher O_2

Description

Players play a 5-on-5 game simulation (figure 23.8).

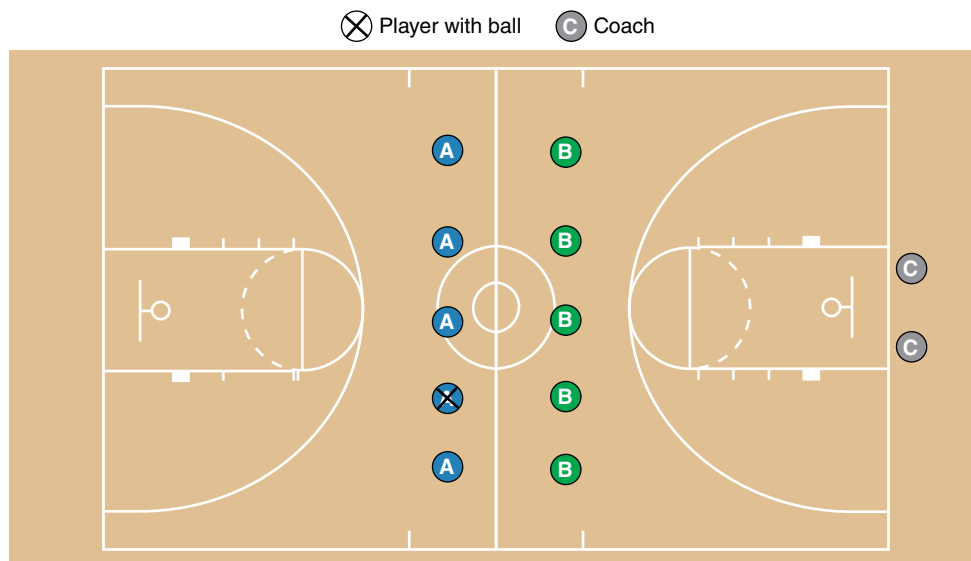


Figure 23.8 5 v 5 scrimmage.

the very-top European teams) consists of 82 games in 25 wk (plus playoff games when applicable), which means an average of 3.5 games per week (with a range of 3-5), leaving the teams little room for practicing, with the competition itself optimizing and maintaining the fitness level of the players. Thus, skill-based conditioning and SSGs will dominate (levels IV and V) in-season. However, we recommend performing level II or level III workouts once a week or every 2 to 3 wk, depending on individual needs (e.g., players not playing, playing less than 15 min, players with body composition goals, or injured players) (44) (see table 23.1 for examples of HIIT types and approaching levels).

Incorporation of HIIT

Ideally, the necessary training is integrated into the basketball sessions themselves, and the drills are designed according to the technical and tactical needs, as well as the physical needs (levels IV and V).

Level III is skill based, usually integrated into the player's individual development sessions with the player development coach and with the presence and support of the strength-and-conditioning coach. Then, unless a player has special needs (e.g., to decrease percentage of body fat, to improve recovery HR, and so on, in which case we may do off-court extra sessions at level 0⁻, 0⁺, I, or II), the HIIT training is part of the basketball practice, where we modulate the load-defining variables, including number of sets, drill duration, work:rest ratios (and rotations), number of teammates and opponents, court size, decision-making complexity, etc.

Sample Training Programs

Table 23.2 provides a typical 3 wk outline of the training routine performed by a professional basketball team, based on the number of games played in a week as well as for different player groups that have different amounts of minutes played per game.

Table 23.2 Sample 21-Day In-Season Schedule, Based on Number of Games per Week and Minutes Played per Game (MP)

This schedule does not show pregame activation or corrective routines, team shoot-around sessions on the morning of game days (no shoot-around during back-to-backs), or strength sessions.

	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Game	Away			Home		Away	Home
< 15 MP	None	SSGs: level III, type 5 and level IV, types 2 and 4	Team practice: level III, type 5 and level IV, types 2 and 4	None	SSGs: level III, type 5 and level IV, types 2 and 4	None	Postgame: level I, types 2 or 3, based on need
15-25 MP	Postgame recovery	Optional skill based 1 v 0 session (level III, ≤ competitive)		Postgame recovery	Recovery	Postgame recovery	Postgame recovery
>25 MP	Postgame recovery	Recovery		Postgame recovery	Recovery	Postgame recovery	Postgame recovery
Game		Home		Home	Away		Home
< 15 MP	Skill-based 1 v 0 session (level III, type 5)	None	SSGs: level III, type 5 and level IV, types 2 and 4	None	Pregame (AM) short skill-based 1 v 0 (level III, type 5)	Day off	None
15-25 MP	Recovery	Postgame recovery	Recovery	Postgame recovery	Postgame recovery	Day off	Postgame recovery
>25 MP	Recovery	Postgame recovery	Recovery	Postgame recovery	Postgame recovery	Day off	Postgame recovery
Game		Home		Home			Away
< 15 MP	SSGs: level III, type 5 and level IV, types 2 and 4	None	SSGs: level III, type 5 and level IV, types 2 and 4	None	Skill-based 1 v 0 session (level III, type 5)	Team practice: level III, type 5 and level 4, types 2 and 4	None
15-25 MP	Recovery	Postgame recovery	Recovery	Postgame recovery	Recovery		Postgame recovery
>25 MP	Recovery	Postgame recovery	Recovery	Postgame recovery	Recovery		Postgame recovery

CASE STUDY

Xavi Schelling

To implement any sort of training methodology as a sports scientist, it is necessary to have complete buy-in from the head coach and the assistant coaches. The development process of the players has to be integrative and oriented to make the player better on court, as well as to maximize the player's durability throughout the season and his or her career. To achieve this buy-in, the proposed methodology must be based on basketball needs. It can't just be focused on physiological principles, but must be aligned with the tactical and technical requirements of the sport.

Each basketball coach has his or her own way of periodizing the implementation of different tactical structures, usually based on a hierarchy of goals. The development of the team's and players' fitness should serve to emphasize these tactical principles as much as possible from a physiological demand perspective while maintaining ecological validity.

The more you get to know the head coach's philosophy, the easier it becomes to periodize and integrate the athletic performance content into the team development program. The following is an example of an integrative methodology I (X. Schelling) experienced, where I integrated conditioning as part of the basketball program.

I was lucky enough to work for 8 yr with the same head coach and friend, Jaume Ponsarnau. We worked together in Bàsquet Manresa (Spanish 1st Division, aka ACB League) and with the Spanish U20 National Team. A mutual understanding allowed us to have a very integrative methodology, by considering each drill as a different aspect of the development process. This meant designing every session together, not considering a type of session to be basketball, conditioning, strength, etc., but more along the lines of what the team needed for that cycle or day or what we wanted to emphasize (i.e., recovery, tactics, conditioning, skills, competition). Each week we had tactical, conditional, and social goals to achieve. Nothing was worked on in isolation, but instead, a continued overlapping of areas was present in each drill.

In 2014, I worked as a sports scientist and strength-and-conditioning coach for the U20 Spanish National team at the European Championship in Crete. We ended up winning the silver medal after a very tight final against Turkey.

The preparation phase with a national team is typically very short (3 or 4 wk). Here we have to make sure the team learns the rules, plays together, and gets in shape. In 2014, we had 3 wk to achieve this. Our main goals in those 3 wk were to maximize the time we spent on court to develop tactical concepts, build team chemistry, and prepare the team to cope with the schedule of the competition (3 games in 3 nights to start the championship).

The key factor we consider when we are preparing for maximal exposure (volume of training load) is taking into account the necessary recovery. To do so, we try to optimize the recovery window (i.e., logistics, treatments, and nutrition), while also managing the intensity of the sessions. In order to quantify the volume and intensity of the sessions, the entire preparation phase was monitored using accelerometers and heart rate monitors (see total acceleration loads per session, in arbitrary units, in figure 23.9).

Conditioning wise, we must keep in mind that the physiological adaptations over 3 wk are going to be limited, and we really rely on the player joining us on the first day with an already developed good level of fitness. Without this, the overload in intensity, frequency, and/or volume could temporarily fatigue the player (chapter 7), which can affect his performance throughout the competition. Thus, the preparation of a national team must start well before the first day of the preparation phase. For this, we communicated with players during the season, and once the season was over, typically 4 to 8 wk before the start of the preparation phase, we contact the players, and their team's strength-and-conditioning coaches, to provide them with the program we want to see that allows the player to recover appropriately from the season but also allows him to prepare for the national team.

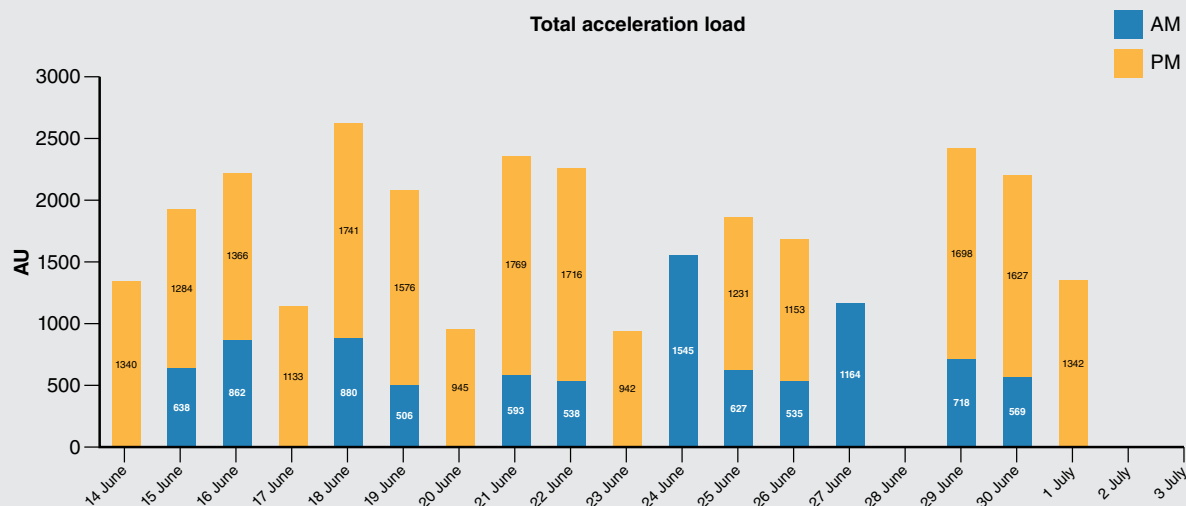


Figure 23.9 Training load as a function of total acceleration load throughout the 3 wk build-up to a major championship (see text for details). AM: morning practice; PM: afternoon practice; AU: arbitrary units.

The program the players completed before joining the team focused on the player's aerobic capacity through long HIIT (levels 0+ and I, type 2) and progressed to develop aerobic power through long- and short-HIIT (level II, types 2 and 3) and skill-based sessions on court (level III, types 4 and 5). The strength program focused on specific weaknesses or injuries the players had during the season (level 0-, 0+, I).

Once the preparation phase in Spain started, we did as much conditioning as we could on court, unless a specific need or injury advised us otherwise (e.g., one of the players came with a mild hamstring strain, and we had him doing an individualized program for 7 d). For the first 3 d of camp, the team progressively increased training volume, but training was without all-out exercises. The drills were mainly half court (level III and IV, types 3 and 4), and on the third day we introduced full-court drills (level IV, types 4 and 5). Since the hardest challenge of the championship was to mentally and physically cope with three straight games of competition, our subcycles within the microcycle were 3 d long, with the fourth day off, emphasizing recovery or very light training (half court with no opposition at low intensity, walk-through or shooting). For the second subcycle of 3 d in the first week, we increased the number of full-court exercises (level IV, types 2-4) and introduced 4 v 4 and 5 v 5 scrimmages (level IV and V, types 2-4).

For the second week, we played a friendly tournament in Spain, which mimicked the 3-games-in-3-nights scenario we knew we'd have to endure (level V). In this tournament, we spread the minutes played evenly among the players, and the stints on court were shorter than the ones we would actually do in the official championship.

In the third week of preparation, we had another friendly tournament (level V), this one in Crete, which also had a 3-in-3 structure. The minutes played and the stints in this event were closer to the ones we had planned to do in the official championship.

Throughout the build-up and competition, we had no injuries to speak of. The talent and desire of the players to provide the best defense in the championship led us to the championship finals after a very tough classification phase. The finals witnessed a beautiful game for all, with good rhythm, tough defense, and a tight score, which, unfortunately for us, ended with a Turkish win. Irrespective of this final score, we had a plan, we executed it as best as we could, and we enjoyed doing it. As one of the assistant coaches said, "Silver medal, golden memories."

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